Innovative Mathematics Olympiad

Model Paper 2025 – 2026

Grade 6

Total Questions: 50 **Time: 1.5 hr.** **Total Marks: 100**

Syllabus

Knowing Our Numbers, Whole Numbers, Playing with Numbers, Basic Geometrical Ideas, Understanding Elementary Shapes, Symmetry, Integers, Fractions and Decimals, Data Handling, Perimeter and Area.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. An app developer wants to create a unique 6-digit ID for users using the digits {0, 1, 2, 5, 8, 9}. The ID cannot start with 0. If A is the largest possible ID and B is the smallest possible ID, what is the value of $A - B$?

(A) 882621 (B) 896301

(C) 887301 (D) 89743

2. A research drone starts at sea level. It descends 15m every minute for 8 minutes to collect samples. It then spots a rare fish and ascends 45 m. After 2 more minutes, it descends another 30 m. What is its final depth represented as an integer?

(A) $-105\ m$ (B) $-135\ m$

(C) $-75\ m$ (D) $-195\ m$

3. **Statement I:** Any number ending in 0 is divisible by 5.
Statement II: Any number divisible by 5 must end in 0.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true and Statement II is false

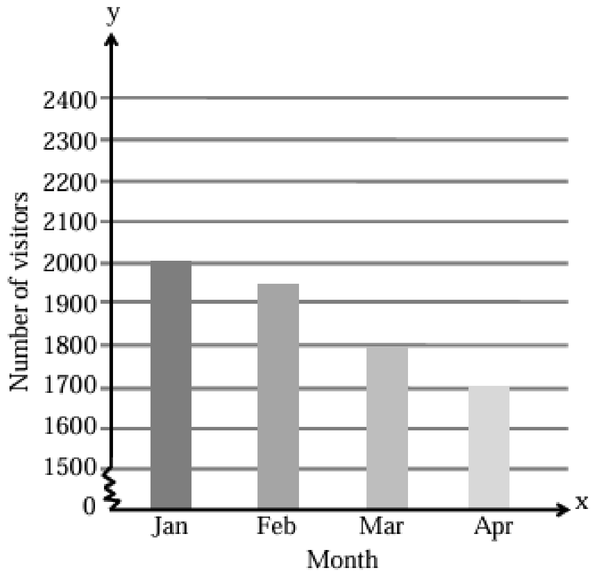
(D) Statement I is false and Statement II is true

4. A room is 6 m long and 4 m wide. A square carpet of side 3 m is laid on the floor. What is the area of the floor that is not carpeted?

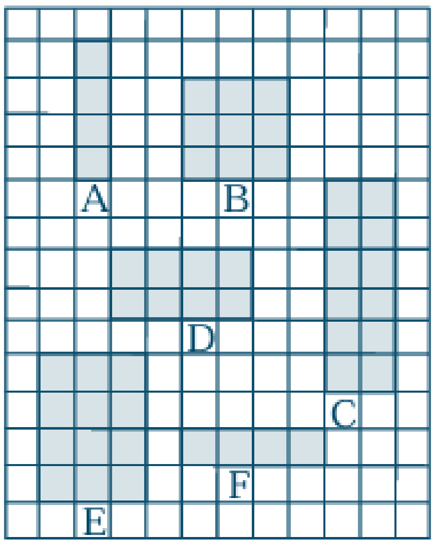
(A) 24 m^2 (B) 9 m^2

(C) 15 m^2 (D) 18 m^2

5. John draws six different rectangles on a centimetre grid paper. Which two



rectangles in the grid have same area?



(A) C and D (B) C and E

(C) B and D (D) A and C

6. Meena wants to buy table cloths for two tables. The respective amounts of cloth required for covering the first and the second tables are 157 cm and 165 cm. What is the total amount of cloth required for covering the two tables?

(A) 2.5 m (B) 2.23 m

(C) 3.22 m (D) 3.4 m

7. What is the correct descending order of the numbers 2.352, 2.253, 2.325, and 2.532?

(A) 2.325 > 2.253 > 2.532 > 2.352

(B) 2.532 > 2.325 > 2.253 > 2.352

(C) 2.532 > 2.352 > 2.325 > 2.253

(D) 2.532 > 2.523 > 2.352 > 2.325

8. The temperature of a town is –14 °C at night. During the day, the temperature increases by 7 °C. What is the temperature of the town during the day?

(A) –7 °C (B) –21 °C

(C) 7 °C (D) –10 °C

9. The given graph shows the number of visitors to an amusement park for four months:

What can be predicted about the number of visitors in May?

(A) They will increase.

(B) They will be the same as in April.

(C) They will decrease.

(D) They will be the same as in February

10. The monthly electricity bills (in ₹) of 25 houses are given below:

266, 261, 264, 275, 262, 272, 274, 276, 261, 276, 273, 282, 280, 282, 272, 264, 266, 278, 279, 280, 271, 282, 269, 266, 275.

The class intervals are constructed as 261-264 (264 not inclusive), 264-267 (267 not inclusive) etc. The interval with maximum frequency is ________.

(A) 261-264 (B) 264-267

(C) 273-276 (D) 282-285

11. The marks (out of 100) obtained by fifty students of a particular class in maths test are given in the frequency table. With respect to the given frequency table

Marks obtained	Number of students
0–10	1
10–20	2
20–30	0
30–40	6
40–50	13
50–60	9
60–70	4
70–80	5
80–90	8
90–100	2
Total	50

some statements are given as:
I. The class interval with highest frequency is 60–70.
II. The class size of each interval is 5.
III. The upper-class limit of the class interval 70–80 is 80.
IV. Two class intervals have the same frequency.
Which two given statements are correct?

(A) I and II (B) III and IV

(C) II and III (D) I and IV

12. Nandini's house is $\frac{9}{10}$ km from her school. She walked some distance and then took a bus for $\frac{1}{2}$ km to reach the school. How far did she walk?

(A) $\frac{2}{5}$ km (B) $\frac{3}{5}$ km

(C) $\frac{7}{5}$ km (D) $\frac{9}{5}$ km

13. T Which of the following fractions is lesser than $\frac{9}{7}$?

(A) $\frac{15}{7}$ (B) $\frac{8}{5}$

(C) $\frac{7}{6}$ (D) $\frac{12}{5}$

14. Which of the following statements is correct?

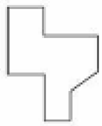
(A) Integers are closed under addition but not under subtraction.

(B) Integers are commutative under subtraction.

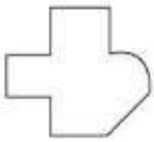
(C) Integers are associative under subtraction.

(D) Integers are commutative under addition.

15. Which of the following figures is not a polygon?

(A)) 

(B) 

(C) 

(D) 

16. The information is given in which alternative is incorrectly matched?

Three-Dimensional Figure	**Number of faces**	**Number of edges**
(A) Rectangular Pyramid	5	8
(B) Triangular Prism	5	9
(C) Cuboid	6	12
(D) Triangular Pyramid	5	9

17. Bryan moves 10 m toward East from a point P and reaches point Q. He then turns left and moves 8 m and reaches point R. From there, he turns left and moves 10 m to reach point S. He again turns left and moves 8 m to reach the starting point (point P). The sum of the angles of the figure enclosed by the Bryan's path is ________.

(A) 720° (B) 620°

(C) 540° (D) 360°

18. A heptagon, hexagon, and octagon can be arranged in the increasing order of their number of their interior angles as ___________.

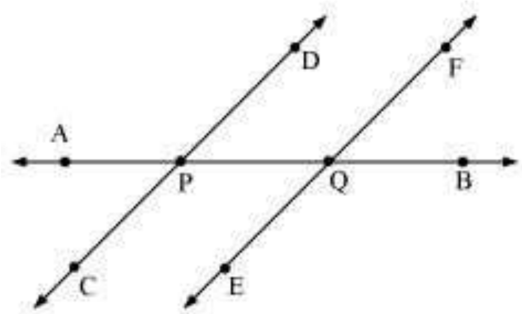
(A) heptagon, hexagon, and octagon

(B) hexagon, heptagon, and octagon

(C) octagon, heptagon, and hexagon

(D) octagon, hexagon, and heptagon

19. The information in which alternative is incorrectly matched?



Angle	**Classification**
(A) ∠CPB	Obtuse
(B) ∠DPQ	Acute
(C) ∠EQA	Acute
(D) ∠AQB	Obtuse

20. What type of angle is made by minute hand and hour hand when time is 7PM?

(A) An acute angle (B) An obtuse angle

(C) A right angle (D) A straight angle

21. If the sum of two numbers is 55 and the HCF and LCM of these numbers are 5 and 120 respectively, then find the sum of the reciprocals of the numbers.

(A) $\frac{55}{600}$ (B) $\frac{110}{120}$

(C) $\frac{11}{12}$ (D) $\frac{220}{240}$

22. If a number is divisible by 30, then which of the following numbers exactly divides the number?

(A) 12 (B) 15

(C) 18 (D) 21

23. The number of consecutive composite numbers less than 100, such that the series contains most numbers, is _____

(A) 6 (B) 5

(C) 8 (D) 7

24. The greatest number which will divide 25, 74 and 99, leaving remainders 1, 2 and 3 respectively is ________.

(A) 17 (B) 18

(C) 24 (D) 22

25. Which of the following alternative is incorrectly matched with respect to the shape arrangement of the corresponding number of dots?

	Number	**Shape**
(A)	3	Triangle
(B)	5	Line
(C)	7	Rectangle
(D)	2	Line

26. Which of the following expressions is an equivalent form of the expression $200 \times 200 - 200 \times 1 + 200 \times 3 - 3 \times 1$?

(A) 203×199 (B) 203×197

(C) 201×199 (D) 201×197

27. What is the difference between the predecessor of 22 and the sum of the successors of the numbers 2971, 926, and 56021?

(A) 59900 (B) 56821

(C) 53300 (D) 51238

28. Which of the following Roman numeral expression is meaningless?
(A) XLIV (B) XCI
(C) IXX (D) XIX

29. There are 6 baskets. Greg puts 6 apples each in 2 baskets and 8 oranges each in the remaining 4 baskets. Total number of fruits in all the baskets is equal to the expression ________
(A) $(6 + 2) \times (8 + 4)$ (B) $(6 \times 2) - (8 + 4)$
(C) $(6 \times 2) + (8 \times 4)$ (D) $(6 + 2) - (8 \times 4)$

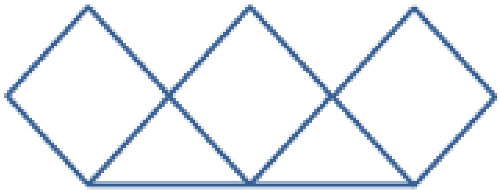
30. Find the greatest four-digit number using the digits 9, 3, 8, and 1 without repetition, such that 1 is at the third place of the number.
(A) 9381 (B) 9183
(C) 9831 (D) 9813

31. Tiya's Beauty Salon charges ₹100 per haircut. If the salon has 72 visitors seeking haircuts per day, estimate how much Tiya would make in a day by rounding the number of visitors to the nearest ten.
(A) ₹ 6,000 (B) ₹ 7,200
(C) ₹ 8,000 (D) ₹ 7,000

32. The smallest number which can be arranged as line, rectangle and triangle is ________
(A) 3 (B) 15
(C) 10 (D) 6

33. Which number should come next in the given series: 7, 10, 15, 22, 31?
(A) 55 (B) 42
(C) 41 (D) 87

34. Find the perimeter of the below given figure, if it is made up of regular polygons only and the length of each side is 9 cm.



(A) 108 cm (B) 140 cm
(C) 126 cm (D) 117 cm

35. Which of the following letters does not have the vertical line of symmetry?
(A) M (B) H
(C) E (D) V

36. Which of the following has an infinite number of lines of symmetry?
(A) Equilateral Triangle
(B) Isosceles Triangle
(C) Regular Hexagon
(D) Circle

37. A rectangular wire of length 40 cm and breadth 20 cm is reshaped into a square. What is it area?
(A) 900 cm^2 (B) 800 cm^2
(C) 120 cm^2 (D) 400 cm^2

38. A bell rings every 18 minutes, a second bell every 24 minutes, and a third bell every 32 minutes. If they all ring together at 8:00 AM, at what time will they next ring together?
(A) 8: 32 AM (B) 10:48 AM
(C) 9: 24 AM (D)12:48 PM

39. Find the value of $(-25) \times [14 + (-18)] - [30 - (-15)]$.
(A) 55 (B) 114
(C) 85 (D) 36

40. In a pie chart representing the expenditure of a family, if the central angle for "Education" is 72°, what percentage of the total income is spent on education?

(A) 15% (B) 25%

(C) 20% (D) 30%

41. The mean of 10 observations was calculated as 40. It was later found that one observation was misread as 35 instead of 53. The correct mean is ____.

(A) 40.8 (B) 41.8

(C) 42.5 (D) 38.2

42. A square and a rectangle have the same perimeter of 80 cm. If the breadth of the rectangle is 4 cm less than the side of the square, find the difference in their areas.

(A) 16 sq.cm (B) 25 sq.cm

(C) 18 sq.cm (D) 20 sq.cm

43. Which of the following figures has rotational symmetry but no line symmetry?

(A) Isosceles triangle (B) Rectangle

(C) Parallelogram (D) Square

44. A rectangular park is surrounded by a path of uniform width 2 m. If the dimensions of the park are 20 m × 12 m, the area of the path is ______

(A) 160 sq.cm (B) 168 sq.cm

(C) 176 sq.cm (D)144sq.cm

45. A figure has rotational symmetry of order 4 and 4 lines of symmetry. Which of the following cannot be that figure?

(A) Square (B) Rhombus

(C) Regular Octagon (D) Both (B) & (C)

46. A deep-sea diver is 150 m below sea level. He descends another 75 m to explore a cave. He then spots a rescue signal and ascends 120 m. If the diver then descends at a rate of 5 m per minute for 10 minutes, what is his final position?

(A) $-155\,m$ (B) $-145\,m$

(C) $-205\,m$ (D) $-105\,m$

47. In a class, $\frac{3}{5}$ of the students are girls and the rest are boys. If $\frac{2}{9}$ of the girls and $\frac{1}{4}$ of the boys are absent, what fraction of the total number of students is present?

(A) $\frac{17}{30}$ (B) $\frac{23}{30}$

(C) $\frac{7}{30}$ (D) $\frac{11}{30}$

48. A drum full of rice weighs 40.16 kg. If the empty drum weighs 13.34 kg, and the rice is distributed into 12 small bags equally, what is the weight of rice in each bag?

(A) 2.235 kg (B) 2.345 kg

(C) 2.135 kg (D) 2.455 kg

49. Find the smallest number which when divided by 20, 25, 35, and 40 leaves remainders 14, 19, 29, and 34 respectively.

(A) 1394 (B) 1406

(C) 1400 (D) 1396

50. A bird is flying at a height of 5000 m above sea level. At a particular point, it is exactly above a submarine floating 1200 m below sea level. The vertical distance between them is ____________

(A) 3800 m (B) 6200 m

(C) 5000 m (D) 1200 m